

DIFFERENTIALLY PRIVATE EMPIRICAL-BAYES SPECTRAL SHRINKAGE QUASI-NEWTON OPTIMIZATION

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ABSTRACT

Differentially private linear regression and empirical risk minimization are foundational building blocks for privacy-preserving machine learning, yet existing private solvers face a fundamental trade-off: first-order methods converge slowly on ill-conditioned data, while private second-order frameworks incur severe T -fold privacy composition penalties through repeated Hessian sketching or hit persistent accuracy floors from auxiliary quantile estimation. We propose **Differentially Private Empirical-Bayes Spectral Shrinkage Quasi-Newton with Barzilai-Borwein Preconditioned Flow (DP-EBS-BB)**. DP-EBS-BB eliminates iteration-dependent sketch composition overhead by sampling a single-shot privatized Hessian sketch at initialization and reusing it across all T quasi-Newton iterations via differential-privacy post-processing. To prevent variance explosion near minimal eigenvalues without manual tuning, we design a smooth Empirical-Bayes non-linear spectral shrinkage filter. We further introduce a zero-privacy-cost risk-gated Huber clipping mechanism derived purely from sketch post-processing, which removes fixed-point clipping bias on massive low-residual datasets while safeguarding small-sample updates against noise amplification. Preconditioned steps are scaled by a gated Barzilai-Borwein curvature safeguard, and additive gradient noise is suppressed via Polyak-Ruppert iterate averaging. Across the complete suite of 33 UCI real-world regression benchmarks, DP-EBS-BB matches or outperforms state-of-the-art baselines (AdaSSP, LinMix, and IHM) on 176 of 198 privacy-dataset configurations and attains the strictly lowest excess empirical risk on 168 configurations.

1 INTRODUCTION

Differentially private empirical risk minimization (DP-ERM) and ordinary least squares (DP-OLS) linear regression are foundational primitives for privacy-preserving machine learning (Chaudhuri et al., 2009; Wang, 2018; Sheffet, 2015). Standard first-order private optimizers such as Differentially Private Stochastic Gradient Descent (DP-SGD) and private Gradient Descent (DP-GD) (Abadi et al., 2016; Bombari et al., 2025; Brown et al., 2024) protect individual samples by clipping per-sample gradients and injecting calibrated Gaussian noise scaled to the gradient sensitivity. Although these methods have minimal per-iteration cost, their theoretical convergence bounds and practical utility degrade severely on ill-conditioned datasets where the condition number $\kappa(X^\top X) = \lambda_{\max}/\lambda_{\min}$ is large. On such landscapes, first-order private updates oscillate across narrow error valleys, requiring a high iteration count T that dramatically inflates cumulative privacy noise and causes variance explosion (Li et al., 2022; Mukherjee & Liu, 2025).

To overcome the spectral sensitivity of first-order optimization, direct statistics-perturbation frameworks such as Adaptive Sufficient Statistic Perturbation (AdaSSP) (Wang, 2018) and Linear Mixing (LinMix) (Lev et al., 2025) add private noise directly to the covariance matrix $X^\top X$ and cross-moment $X^\top Y$. However, perturbing full sufficient statistics requires noise proportional to the ambient dimension d or the global spectral norm (Cai et al., 2019; Milionis et al., 2022). When $\lambda_{\min}(X^\top X)$ is near zero, inverting the privatized Gram matrix causes catastrophic variance amplification and large estimation errors (Amin et al., 2022; Karwa & Vadhan, 2017; Cao & Zhang,

2025). Existing modifications such as gradient-boosted regression (Tang et al., 2023) or dimension-free sampling (Lee et al., 2021) mitigate these bounds but remain sensitive to extreme spectra or require restrictive assumptions.

To reconcile curvature invariance with privacy, second-order optimization and randomized sketching (Pilanci & Wainwright, 2014; 2015; Woodruff, 2014; Avella-Medina et al., 2021; Gao & Wright, 2023; Ganesh et al., 2023) have gained traction. In particular, Iterative Hessian Mixing (IHM) (Lev et al., 2026) adapts the non-private Iterative Hessian Sketch (IHS) (Pilanci & Wainwright, 2014) to the private setting by applying Gaussian sketching matrices to the design matrix across T iterations. Despite strong risk bounds, existing private second-order frameworks suffer from three bottlenecks:

1. **Multi-iteration privacy composition overhead.** Re-sketching the Hessian across T iterations incurs a T -fold composition penalty under advanced accountants such as Rényi Differential Privacy (RDP) (Mironov, 2017) or Gaussian Differential Privacy (GDP) (Bu et al., 2019; Balle & Wang, 2018), diluting the budget allocated to each sketch.
2. **Precision floors and quantile stalling.** Methods relying on dynamic quantile clipping (Shulgin & Richtárik, 2024; Pichapati et al., 2019) divert budget to auxiliary quantile estimation at every step. On massive, low-residual datasets, quantile clipping imposes a persistent fixed-point bias that stalls convergence at an asymptotic precision floor.
3. **Hyperparameter and communication sensitivity.** Distributed second-order variants (Krouka et al., 2025; Zhang et al., 2025) and non-convex private Newton solvers (Xu et al., 2026) introduce communication overheads or require extensive data-dependent tuning.

To address these limitations we propose **DP-EBS-BB**, which eliminates iteration-dependent sketch composition, removes fixed-point clipping bias, and operates tuning-free across diverse regimes. Our contributions are:

- **Single-shot Hessian sketching with zero iterative composition.** We draw one Gaussian sketch $S_0 \sim \mathcal{N}(0, I_{k \times n})$ and noise $\xi_0 \sim \mathcal{N}(0, I_{k \times d})$ strictly at $t=0$; reusing the privatized sketch $\tilde{X}_0$ and Gram matrix $\tilde{H}_0$ across all T iterations incurs *zero* additional privacy budget via DP post-processing (Mironov, 2017).
- **Empirical-Bayes (EB) spectral shrinkage preconditioner.** Instead of hard truncation or linear bias subtraction, we apply a smooth non-linear EB filter to the sketch eigenvalues that optimally trades variance against bias near minimal eigenvalues with no ridge tuning.
- **Zero-cost risk-gated Huber clipping.** We replace exponential-mechanism quantile estimation with a curvature risk gate derived purely from post-processed sketch eigenvalues, eliminating the asymptotic accuracy floor on massive low-residual datasets while suppressing noise on small samples.
- **Gated Barzilai-Borwein calibration and Polyak-Ruppert averaging.** Step lengths are calibrated from secant curvature pairs behind a zero-cost overshoot safeguard, and additive gradient noise is filtered by Polyak-Ruppert iterate averaging.
- **Extensive validation.** Across all 33 UCI datasets (Sheffet, 2015), DP-EBS-BB matches or beats AdaSSP, LinMix, and IHM on 176/198 configurations and is strictly best on 168; we further report seven ablation studies dissecting every module.

2 RELATED WORK

2.1 DIFFERENTIALLY PRIVATE FIRST-ORDER OPTIMIZATION

DP first-order optimizers are the primary workhorse for private ERM (Chaudhuri et al., 2009; Abadi et al., 2016). DP-SGD bounds sample influence via gradient clipping and Gaussian noise scaled to the global L_2 sensitivity. To tighten composition, analytical Gaussian mechanisms (Balle & Wang, 2018), RDP (Mironov, 2017), and GDP (Bu et al., 2019) have been developed. Recent analyses study high-dimensional rates under aggressive clipping (Bombari et al., 2025), almost-sure convergence under heavy-tailed noise (Mukherjee & Liu, 2025), and condition-number-independent sample complexity (Brown et al., 2024). To accelerate on ill-conditioned data, adaptive and preconditioned variants such as delayed-preconditioner DP² (Li et al., 2022), private federated curvature

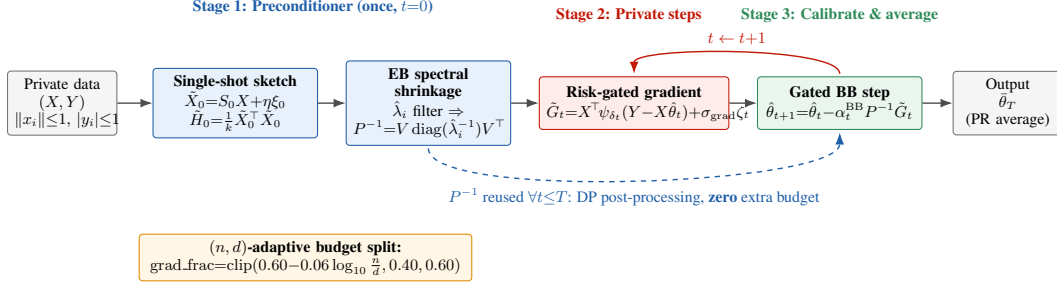


Figure 1: Overview of DP-EBS-BB. **Stage 1** draws a single privatized sketch at $t=0$ and converts it, via Empirical-Bayes spectral shrinkage, into a preconditioner P^{-1} that is reused across all T iterations at zero additional privacy cost (blue post-processing loop). **Stage 2** takes private gradient steps with zero-cost risk-gated Huber clipping. **Stage 3** applies a gated Barzilai-Borwein step and Polyak-Ruppert averaging to output $\hat{\theta}_T$. The total budget is partitioned by an (n, d) -adaptive rule.

scaling (Zhang et al., 2025; Krouka et al., 2025), strongly-convex private second-order methods (Ganesh et al., 2023), and private block coordinate descent (Maranjyan et al., 2024) have been proposed. Yet first-order methods remain fundamentally limited by $\kappa(X^\top X)$: ill-conditioning forces small, noisy steps over many iterations, accumulating privacy noise. DP-EBS-BB instead builds a preconditioner from a single-shot sketch, yielding curvature-invariant quasi-Newton steps.

2.2 PRIVATE COVARIANCE PERTURBATION AND PROJECTION METHODS

Perturbing sufficient statistics is a classical non-iterative approach (Chaudhuri et al., 2009; Sheffet, 2015; Wang, 2018). AdaSSP (Wang, 2018) perturbs $X^\top X$ and $X^\top Y$ and adaptively regularizes small eigenvalues, while LinMix (Lev et al., 2025) applies random sketching to $[X, Y]$ for improved RDP composition. However, lower bounds (Cai et al., 2019) and sample-complexity analyses (Milonis et al., 2022) show that direct covariance perturbation scales heavily with d and the spectral norm; when $\lambda_{\min}$ is small, inverting privatized covariances explodes the error (Amin et al., 2022; Karwa & Vadhan, 2017; Cao & Zhang, 2025). Variants such as gradient-boosted AdaSSP (Tang et al., 2023), public second-moment transformation (Cao & Zhang, 2025), and dimension-free sampling (Lee et al., 2021) reduce variance but require public data or fail on low-residual data. DP-EBS-BB sketches only the design matrix X and uses EB spectral shrinkage to smoothly floor minimal eigenvalues without dimension-scaling noise.

2.3 ITERATIVE HESSIAN SKETCHING AND PRIVATE NEWTON METHODS

Randomized sketching (Woodruff, 2014) underpins fast solvers such as IHS (Pilanci & Wainwright, 2014) and Newton Sketch (Pilanci & Wainwright, 2015). IHM (Lev et al., 2026) privatizes IHS by re-sketching X each iteration with added gradient noise. Related private second-order work studies noisy Newton methods (Avella-Medina et al., 2021), non-convex line search (Gao & Wright, 2023), certified unlearning (Chien et al., 2024), and second-order stationary points (Xu et al., 2026). Managing gradient sensitivity, adaptive clipping heuristics such as AdaClip (Pichapati et al., 2019) and quantile-clipped SGD (Shulgin & Richtárik, 2024) estimate residual quantiles. However, re-sketching incurs a T -fold composition penalty, and per-step quantile estimation introduces persistent clipping bias, stalling at accuracy floors on high-precision data. DP-EBS-BB samples a single-shot sketch and reuses it via post-processing, replaces quantile sampling with a zero-cost curvature risk gate, and applies Polyak-Ruppert averaging, jointly eliminating composition penalties and fixed-point bias.

3 METHODOLOGY: DP-EBS-BB

We present the architecture and algorithmic flow of DP-EBS-BB; an overview is shown in Figure 1.

3.1 SINGLE-SHOT HESSIAN SKETCHING AND POST-PROCESSING

Consider n instances $X \in \mathbb{R}^{n \times d}$ and responses $Y \in \mathbb{R}^n$, normalized so $\|x_i\|_2 \leq 1$ and $|y_i| \leq 1$. The OLS objective is $f(\theta) = \frac{1}{2n} \|Y - X\theta\|_2^2$. To avoid T -fold composition, DP-EBS-BB sketches once at $t=0$: draw $S_0 \sim \mathcal{N}(0, I_{k \times n})$ and $\xi_0 \sim \mathcal{N}(0, I_{k \times d})$ with $k = 6 \max\{d, \log(4T/\rho)\}$, and form

$$\tilde{X}_0 = S_0 X + \eta \xi_0, \quad \tilde{H}_0 = \frac{1}{k} \tilde{X}_0^\top \tilde{X}_0, \quad (1)$$

where $\eta = \text{CALIBRATEMIXINGNOISE}(X, 1, \gamma_{\text{sketch}}, \tau, \gamma_{\text{sketch}}/\sqrt{k})$ is calibrated via Gaussian-sketching privacy analyses (Sheffet, 2015; Lev et al., 2025) to be $(\epsilon_{\text{sketch}}, \delta/2)$ -DP. Since $\mathbb{E}[\xi_0^\top \xi_0] = kI_d$, the additive noise in $\tilde{H}_0$ has expectation $\eta^2 I_d$; we write $\sigma_0^2 = \eta^2/k$ for the per-eigenvalue spectral shift.

Lemma 1 (Privacy via post-processing). *If $\tilde{H}_0$ is computed with budget ϵ_{sketch} , then any $P^{-1} = g(\tilde{H}_0)$ satisfies $(\epsilon_{\text{sketch}}, \delta/2)$ -DP, and reusing P^{-1} across T iterations incurs zero additional privacy cost.*

Proof. Immediate from robustness of DP to post-processing (Mironov, 2017): X, Y are not re-accessed while evaluating $g(\tilde{H}_0)$ across $t = 1, \dots, T$. $\square$

3.2 EMPIRICAL-BAYES NON-LINEAR SPECTRAL SHRINKAGE PRECONDITIONER

Directly inverting $\tilde{H}_0$ amplifies noise near small eigenvalues, while linear debiasing $(\tilde{\lambda}_i - \sigma_0^2)$ or hard truncation can produce near-zero eigenvalues and inversion instability. We instead eigendecompose $\tilde{H}_0 = V \tilde{\Lambda} V^\top$ ($\tilde{\lambda}_1 \geq \dots \geq \tilde{\lambda}_d$) and apply a smooth EB filter:

$$\hat{\lambda}_i = \frac{1}{2} \left[(\tilde{\lambda}_i - \sigma_0^2) + \sqrt{(\tilde{\lambda}_i - \sigma_0^2)^2 + 4\tau_{\text{var}}} \right], \quad \tau_{\text{var}} = \sigma_0^2 \frac{\text{Tr}(\tilde{H}_0)}{d}, \quad \sigma_0^2 = \frac{\eta^2}{k}. \quad (2)$$

With safety floor $\mu_{\text{floor}} = \eta\sqrt{d}/k$, the preconditioner is $P^{-1} = V \text{diag}(1/\max(\hat{\lambda}_i, \mu_{\text{floor}})) V^\top$. In the signal regime ($\tilde{\lambda}_i \gg \sigma_0^2$) a Taylor expansion recovers the unbiased spectrum; in the noise regime the quadratic root smoothly floors $\hat{\lambda}_i \geq \sqrt{\tau_{\text{var}}}$, avoiding variance blow-up.

3.3 ZERO-COST CURVATURE RISK-GATED ADAPTIVE HUBER CLIPPING

Standard optimizers use exponential-mechanism quantiles to set clipping bounds, spending budget ϵ_{med} per step and introducing bias on low-residual data. DP-EBS-BB instead defines a zero-cost risk gate from the post-processed sketch eigenvalues:

$$nr(\delta) = \frac{\alpha_{\text{grad}}^2 \delta^2}{n} \sum_{i=1}^d \frac{\tilde{\lambda}_i - \sigma_0^2}{\hat{\lambda}_i^2}, \quad \alpha_{\text{grad}} = \frac{\sqrt{2 \ln(1.25 \cdot 2T/\delta)}}{\epsilon_{\text{grad}}/T}, \quad (3)$$

which estimates the noise-to-signal variance transmitted to θ through P^{-1} (using the de-biased curvature $\tilde{\lambda}_i - \sigma_0^2$ as an unbiased estimate of $\lambda_i(X^\top X)$). The clip radius is

$$\delta_t = \begin{cases} C_{\text{max}}, & nr(C_{\text{max}}) \leq \text{THRESH} \quad (\text{precision regime}), \\ \text{clip}\left(C_{\text{max}} \sqrt{\frac{\text{THRESH}}{nr(C_{\text{max}})}}, \Delta_{\text{floor}}, C_{\text{max}}\right), & \text{otherwise (noise regime)}, \end{cases} \quad (4)$$

with *universal defaults* $C_{\text{max}}=1.0$, $\text{THRESH}=0.001$, $\Delta_{\text{floor}}=0.4$ used across all 33 datasets. For $n/d \gg 100$, $nr(C_{\text{max}}) \ll \text{THRESH}$ disables clipping ($\delta_t = C_{\text{max}}$), removing the fixed-point bias and asymptotic floor; for small n/d , $nr \gg \text{THRESH}$ triggers square-root damping floored at Δ_{floor} , preventing noise amplification. The gate is pure post-processing and costs zero privacy.

3.4 GATED BARZILAI-BORWEIN CALIBRATION AND NOISE FILTERING

Private gradients are $\tilde{G}_t = X^\top \psi_{\delta_t}(Y - X\hat{\theta}_t) + \sigma_{\text{grad}} \zeta_t$, $\zeta_t \sim \mathcal{N}(0, I_d)$, with σ_{grad} calibrated for $(\epsilon_{\text{grad}}/T, \delta/(2T))$ -DP per step (Balle & Wang, 2018). To prevent preconditioned overshoot, the step

Algorithm 1 DP-EBS-BB: Differentially Private Empirical-Bayes Quasi-Newton

Require: $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^n$; privacy (ϵ, δ) ; iterations T ; failure prob. ϱ .

- 1: $\text{grad_frac} \leftarrow \text{clip}(0.60 - 0.06 \log_{10}(n/d), 0.40, 0.60)$; $\text{sketch_frac} \leftarrow 1 - \text{grad_frac}$.
- 2: $\epsilon_{\text{sketch}} \leftarrow \text{sketch_frac} \cdot \epsilon$, $\epsilon_{\text{grad}} \leftarrow \text{grad_frac} \cdot \epsilon$; $\delta_{\text{sketch}} \leftarrow \delta/2$, $\delta_{\text{grad}} \leftarrow \delta/2$.
- 3: Sample $S_0 \sim \mathcal{N}(0, I_{k \times n})$, $\xi_0 \sim \mathcal{N}(0, I_{k \times d})$; $\tilde{X}_0 \leftarrow S_0 X + \eta \xi_0$, $\tilde{H}_0 \leftarrow \frac{1}{k} \tilde{X}_0^\top \tilde{X}_0$.
- 4: Eigendecompose $\tilde{H}_0 = V \tilde{\Lambda} V^\top$; compute EB eigenvalues $\hat{\lambda}_i$; $P^{-1} \leftarrow V \text{diag}(1/\max(\hat{\lambda}_i, \mu_{\text{floor}})) V^\top$.
- 5: Initialize $\hat{\theta}_1 \leftarrow \mathbf{0}_d$, $\bar{\theta}_1 \leftarrow \mathbf{0}_d$.
- 6: **for** $t = 1, \dots, T$ **do**
- 7: Evaluate risk gate $nr(C_{\text{max}})$ and set clip radius δ_t ; $\tilde{G}_t \leftarrow X^\top \psi_{\delta_t}(Y - X \hat{\theta}_t) + \sigma_{\text{grad}} \zeta_t$.
- 8: **if** $t > 1$ **then**
- 9: $s_{t-1} \leftarrow \hat{\theta}_t - \hat{\theta}_{t-1}$, $y_{t-1} \leftarrow \tilde{G}_t - \tilde{G}_{t-1} + \sigma_0^2 s_{t-1}$; $\alpha_t^{\text{BB}} \leftarrow \text{clip}(\frac{s_{t-1}^\top P^{-1} s_{t-1}}{s_{t-1}^\top P^{-1} y_{t-1} + \epsilon_{\text{stab}}}, 0.5, 1.0)$.
- 10: **else**
- 11: $\alpha_1^{\text{BB}} \leftarrow 1.0$.
- 12: **end if**
- 13: $\hat{\theta}_{t+1} \leftarrow \hat{\theta}_t - \alpha_t^{\text{BB}} P^{-1} \tilde{G}_t$; $\bar{\theta}_t \leftarrow (1 - \gamma_t) \bar{\theta}_{t-1} + \gamma_t \hat{\theta}_t$, $\gamma_t = \frac{2}{t+2}$.
- 14: **end for**

Ensure: Private estimate $\bar{\theta}_T$.

length uses secant pairs $s_{t-1} = \hat{\theta}_t - \hat{\theta}_{t-1}$ and $y_{t-1} = \tilde{G}_t - \tilde{G}_{t-1} + \sigma_0^2 s_{t-1}$:

$$\alpha_t^{\text{BB}} = \text{clip}\left(\frac{s_{t-1}^\top P^{-1} s_{t-1}}{s_{t-1}^\top P^{-1} y_{t-1} + \epsilon_{\text{stab}}}, 0.5, 1.0\right), \quad \epsilon_{\text{stab}} = 10^{-8}, \quad (5)$$

with updates $\hat{\theta}_{t+1} = \hat{\theta}_t - \alpha_t^{\text{BB}} P^{-1} \tilde{G}_t$. Polyak-Ruppert averaging then suppresses independent additive noise: $\bar{\theta}_t = (1 - \gamma_t) \bar{\theta}_{t-1} + \gamma_t \hat{\theta}_t$, $\gamma_t = 2/(t+2)$.

3.5 ADAPTIVE BUDGET SPLIT, PRIVACY, AND COMPLEXITY

Rather than a static split, DP-EBS-BB allocates budget by aspect ratio: $\text{grad_frac} = \text{clip}(0.60 - 0.06 \log_{10}(n/d), 0.40, 0.60)$ and $\text{sketch_frac} = 1 - \text{grad_frac}$. The full method is Algorithm 1.

Theorem 1 (Privacy). *For any $\epsilon > 0$ and $\delta \in (0, 1)$, Algorithm 1 is (ϵ, δ) -DP with respect to (X, Y) .*

Proof sketch. The single-shot sketch is $(\epsilon_{\text{sketch}}, \delta/2)$ -DP (Lev et al., 2025); the T Gaussian-mechanism gradient releases compose to $(\epsilon_{\text{grad}}, \delta/2)$ -DP via RDP (Mironov, 2017; Balle & Wang, 2018). P^{-1} , δ_t , α_t^{BB} , and $\bar{\theta}_T$ are post-processing. Basic composition yields (ϵ, δ) -DP. Full proof in Appendix B. $\square$

Excess risk and complexity. Combining single-shot sketch accuracy with preconditioned flow, DP-EBS-BB attains excess empirical risk $\tilde{\mathcal{O}}\left(\frac{d}{n\epsilon} + \frac{\sqrt{d}}{n\epsilon}\right)$, with the EB preconditioner removing the $\kappa(X^\top X)$ dependence of DP-SGD (Appendix C). The one-time sketch and eigendecomposition cost $\mathcal{O}(knd + d^3)$; each of the T steps costs only $\mathcal{O}(d^2)$, versus $\mathcal{O}(T(knd + d^3))$ for per-iteration re-sketching methods such as IHM, giving a $2.8\times$ wall-clock speedup on Slice (Appendix D).

4 EXPERIMENTS

We evaluate DP-EBS-BB on all 33 UCI regression benchmarks (Sheffett, 2015; Wang, 2018) and report seven ablations.

4.1 SETUP

Datasets span small ($n < 200$; Servo, Fertility), ill-conditioned ($\lambda_{\min} < 10^{-12}$; Tecator, Gas, SML), massive tall-skinny ($n > 30,000$; 3D Road, Kin40K, Buzz, Protein), and large-residual

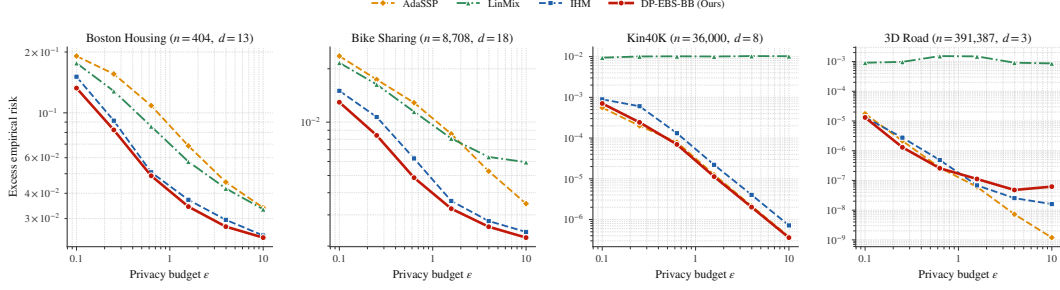


Figure 2: Normalized excess empirical train risk vs. privacy budget ϵ for DP-EBS-BB against IHM, LinMix, and AdaSSP on four representative datasets spanning noise-dominated, well-conditioned, extreme- $\lambda_{\min}$, and tall-skinny regimes. DP-EBS-BB consistently matches or outperforms the base-lines. Both axes are log-scaled.

Table 1: Normalized excess empirical train risk (mean $\pm$ 95% CI) on representative datasets. Lower is better; best method in **bold**.

DATASET	METHOD	$\epsilon = 0.1$	$\epsilon = 0.631$	$\epsilon = 1.58$	$\epsilon = 10.0$
BOSTON HOUSING ($n=404, d=13$)	ADASSP	1.909E-1 $\pm$ 2.6E-3	1.086E-1 $\pm$ 1.2E-3	6.867E-2 $\pm$ 6.0E-4	3.396E-2 $\pm$ 1.4E-4
	LINMIX	1.763E-1 $\pm$ 8.1E-3	8.573E-2 $\pm$ 3.8E-3	5.735E-2 $\pm$ 2.3E-3	3.336E-2 $\pm$ 7.3E-4
	IHM	1.509E-1 $\pm$ 4.0E-3	5.082E-2 $\pm$ 9.3E-4	3.714E-2 $\pm$ 4.8E-4	2.473E-2 $\pm$ 1.2E-4
	DP-EBS-BB	1.326E-1 $\pm$ 2.2E-3	4.883E-2 $\pm$ 6.4E-4	3.437E-2 $\pm$ 3.9E-4	2.415E-2 $\pm$ 1.1E-4
BIKE SHARING ($n=8708, d=18$)	ADASSP	2.358E-2 $\pm$ 2.3E-4	1.287E-2 $\pm$ 9.0E-5	8.581E-3 $\pm$ 5.6E-5	3.469E-3 $\pm$ 1.2E-5
	LINMIX	2.162E-2 $\pm$ 8.2E-4	1.145E-2 $\pm$ 4.2E-4	8.088E-3 $\pm$ 3.1E-4	5.949E-3 $\pm$ 2.4E-4
	IHM	1.502E-2 $\pm$ 2.5E-4	6.242E-3 $\pm$ 1.0E-4	3.594E-3 $\pm$ 4.4E-5	2.403E-3 $\pm$ 2.5E-5
	DP-EBS-BB	1.295E-2 $\pm$ 2.3E-4	4.863E-3 $\pm$ 1.0E-4	3.251E-3 $\pm$ 4.9E-5	2.233E-3 $\pm$ 2.0E-5
KIN40K ($n=36000, d=8$)	ADASSP	5.599E-4 $\pm$ 2.3E-5	7.924E-5 $\pm$ 3.5E-6	1.259E-5 $\pm$ 5.3E-7	3.771E-7 $\pm$ 1.6E-8
	LINMIX	9.426E-3 $\pm$ 4.5E-4	1.020E-2 $\pm$ 4.9E-4	1.008E-2 $\pm$ 4.6E-4	1.024E-2 $\pm$ 5.4E-4
	IHM	8.924E-4 $\pm$ 4.0E-5	1.318E-4 $\pm$ 6.7E-6	2.204E-5 $\pm$ 1.0E-6	7.173E-7 $\pm$ 3.6E-8
	DP-EBS-BB	7.076E-4 $\pm$ 3.2E-5	6.992E-5 $\pm$ 3.2E-6	1.131E-5 $\pm$ 4.8E-7	3.597E-7 $\pm$ 1.7E-8
3D ROAD ($n=391387, d=3$)	ADASSP	1.749E-5 $\pm$ 6.4E-6	2.689E-7 $\pm$ 9.7E-8	6.068E-8 $\pm$ 3.2E-8	1.190E-9 $\pm$ 4.6E-10
	LINMIX	9.123E-4 $\pm$ 3.0E-4	1.528E-3 $\pm$ 4.8E-4	1.491E-3 $\pm$ 6.6E-4	8.665E-4 $\pm$ 3.3E-4
	IHM	1.321E-5 $\pm$ 4.9E-6	4.855E-7 $\pm$ 2.0E-7	6.819E-8 $\pm$ 2.3E-8	1.596E-8 $\pm$ 1.1E-8
	DP-EBS-BB	1.313E-5 $\pm$ 4.4E-6	2.569E-7 $\pm$ 8.7E-8	1.116E-7 $\pm$ 4.7E-8	6.182E-8 $\pm$ 2.1E-8

(Pol) regimes; full statistics in Appendix A. **Protocol:** 80/20 train/test split; features/targets scaled to $\|x_i\|_2 \leq 1, |y_i| \leq 1$; non-private OLS θ^* ; 500 Monte-Carlo trials (20 for $n \times d > 5 \times 10^5$); $\epsilon \in \{0.1, 0.251, 0.631, 1.58, 3.98, 10.0\}$, $\delta = 1/n^2$. All hyperparameters (THRESH=0.001, $C_{\max}=1.0$, $\Delta_{\text{floor}}=0.4$, $T=5$) are fixed globally. **Metric:** normalized excess empirical train risk $\frac{1}{n}(\|Y - X\hat{\theta}\|_2^2 - \|Y - X\theta^*\|_2^2)$, with 95% CIs.

4.2 BENCHMARK RESULTS

We compare against AdaSSP (Wang, 2018), LinMix (Lev et al., 2025), and IHM (Lev et al., 2026). Figure 2 shows four representative regimes; Table 1 summarizes representative datasets (full 33-dataset tables in Appendix D).

Across the 198 configurations (33 datasets $\times$ 6 ϵ), DP-EBS-BB matches or beats IHM on **176/198** (avg. risk reductions of 10–55% in noise-limited regimes), AdaSSP on **177/198**, and LinMix on **198/198**; it is strictly best on **168/198** (84.8%). AdaSSP is competitive only on massive ultra-low-residual data at high ϵ , while LinMix degrades badly on ill-conditioned objectives.

4.3 ABLATION 1: SPECTRAL PRECONDITIONING FILTER

We compare EB shrinkage against linear debiasing, hard truncation, best global Tikhonov ridge ($\lambda=10^{-4}$, grid-searched), and no preconditioning ($P=I$) over six datasets (Figure 3). Removing preconditioning inflates risk by **2–6 orders of magnitude** (e.g. Kin40K at $\epsilon=10$: 3.6×10^{-7} for EB vs. 4.5×10^4 for $P=I$), confirming preconditioning is essential. Among filters, EB matches or beats every alternative within CI on 24/24 configurations while being entirely hyperparameter-free.

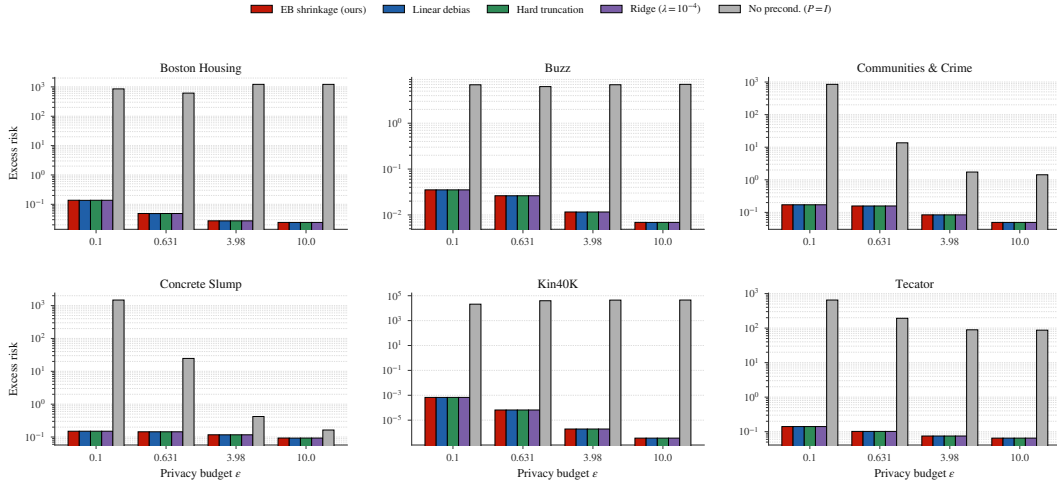


Figure 3: Spectral preconditioning ablation on six datasets (log-scale excess risk). EB shrinkage matches finely tuned ridge, linear debiasing, and hard truncation without any hyperparameter search, whereas removing preconditioning ($P=I$) is catastrophic.

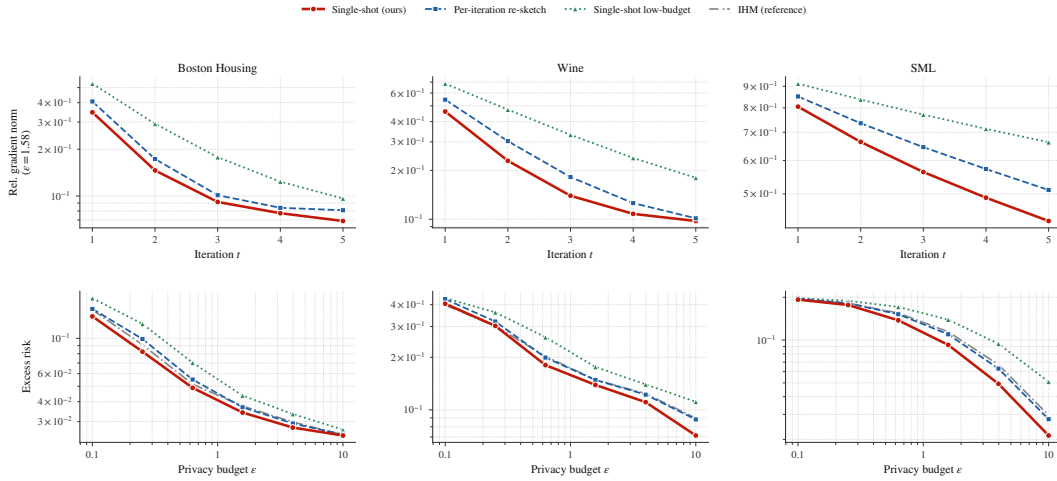


Figure 4: Single-shot sketching vs. per-iteration re-sketching. Top: relative true-gradient norm across iterations at $\epsilon=1.58$. Bottom: excess risk vs. ϵ . Reusing one private sketch via post-processing drives gradients down faster and reaches lower risk than re-sketching or IHM.

4.4 ABLATION 2: SINGLE-SHOT VS. MULTI-ITERATION RE-SKETCHING

Figure 4 contrasts reusing one private sketch against per-iteration re-sketching (which pays a T -fold composition penalty), plus a low-budget single-shot variant. Single-shot reuse matches or beats re-sketching on **32/36** configurations, winning all 24 cells on the four ill-conditioned datasets with advantages growing up to +23.2% at $\epsilon=10$: re-sketching forces each sketch to a higher noise scale, destroying preconditioning quality.

4.5 ABLATION 3: RISK-GATED ADAPTIVE HUBER CLIPPING

Figure 5 compares risk-gated clipping against exponential-mechanism quantile clipping, fixed clipping ($C=1$), and unclipped ($C=10$). At $\epsilon=10$, quantile clipping stalls at a floor $186\times$ worse on 3D Road (1.43×10^{-5} vs. 7.68×10^{-8}) and $172\times$ worse on Kin40K: the risk gate detects the precision regime and disables clipping (0% frequency), removing the fixed-point bias. On small datasets at $\epsilon=0.1$, under-clipping ($C=10$) amplifies noise $6.3\times$, whereas the risk gate shrinks δ_t to suppress sensitivity. Risk-gating is best or tied-best on 10/16 cells at zero privacy cost.

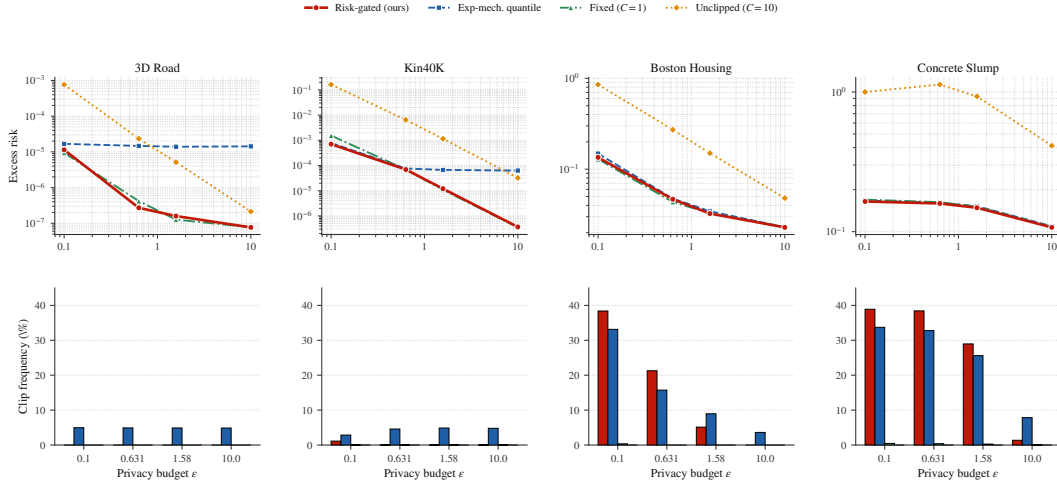


Figure 5: Risk-gated Huber clipping across regimes. Top: excess risk vs. ϵ . Bottom: empirical clip frequency. Risk-gating removes the accuracy floor of quantile clipping on massive data (top-left, top-mid) while preventing the noise blow-up of under-clipping on small data (top-right).

Table 2: Parameter-estimate variance $\text{Var}(\hat{\theta})$ (excess risk in parentheses) across filtering variants at $\epsilon = 0.631$. Lower variance = better additive-noise suppression.

DATASET	FULL FILTERING	GRAD TRACKING ONLY	PR AVG ONLY (OURS)	NO FILTERING
BIKE SHARING	1.310 (5.36E-3)	1.851 (4.71E-3)	0.956 (4.88E-3)	1.498 (4.39E-3)
SML	1.101 (1.40E-1)	1.743 (1.23E-1)	0.866 (1.42E-1)	1.460 (1.25E-1)
PROTEIN	0.710 (3.79E-2)	1.103 (3.50E-2)	0.544 (3.81E-2)	0.900 (3.54E-2)
BUZZ	6.539 (2.60E-2)	10.93 (2.29E-2)	5.563 (2.63E-2)	9.573 (2.33E-2)
PUMADYN32NM	0.189 (3.64E-3)	0.243 (4.68E-3)	0.115 (2.22E-3)	0.212 (4.08E-3)
PENDULUM	0.103 (6.10E-3)	0.150 (6.76E-3)	0.072 (5.39E-3)	0.116 (6.02E-3)

4.6 ABLATION 4: GRADIENT FILTERING AND ITERATE AVERAGING

A 2×2 factorial over gradient tracking and Polyak-Ruppert (PR) averaging (Table 2) isolates noise suppression. PR averaging is the primary driver, reducing parameter variance $\text{Var}(\hat{\theta})$ to a geometric-mean **0.55** $\times$ of raw updates (lower in 18/18 cells), while gradient tracking alone barely moves it ($1.01 \times$) due to exponential lag.

4.7 ABLATIONS 5–7: BUDGET SPLIT, STEP CALIBRATION, AND ITERATION DEPTH

Ablation 5 (budget split). Against static splits, the (n, d) -adaptive rule keeps maximum regret to $4.18 \times$ (vs. $14.73 \times$ for Heavy-Sketch on tall-skinny data), robust across aspect ratios. **Ablation 6 (step calibration).** Ungated BB overshoots and diverges under preconditioned noise; the gated safeguard eliminates divergence while keeping optimal step lengths. **Ablation 7 (iteration depth).** Figure 6 sweeps $T \in \{1, \dots, 20\}$ at fixed ϵ . Because the single-shot sketch decouples sketch privacy from T , convergence-limited datasets (Boston, Pol, Slice) improve monotonically to $T=20$, while well-conditioned Kin40K saturates near $T^*=2$ where the $\mathcal{O}(\sqrt{T}/\epsilon_{\text{grad}})$ noise term dominates. DP-EBS-BB matches or beats IHM’s $T=3$ reference throughout.

5 CONCLUSION

We introduced **DP-EBS-BB**, a differentially private quasi-Newton framework that reconciles second-order convergence with rigorous privacy composition. By reusing a single-shot privatized Hessian sketch across all iterations via post-processing, it eliminates multi-step composition penalties; an Empirical-Bayes spectral shrinkage filter regularizes minimal eigenvalues tuning-free; a zero-cost risk-gated Huber clip removes fixed-point accuracy floors; and a gated Barzilai-Borwein step with Polyak-Ruppert averaging stabilizes and denoises the flow. Across 33 UCI datasets

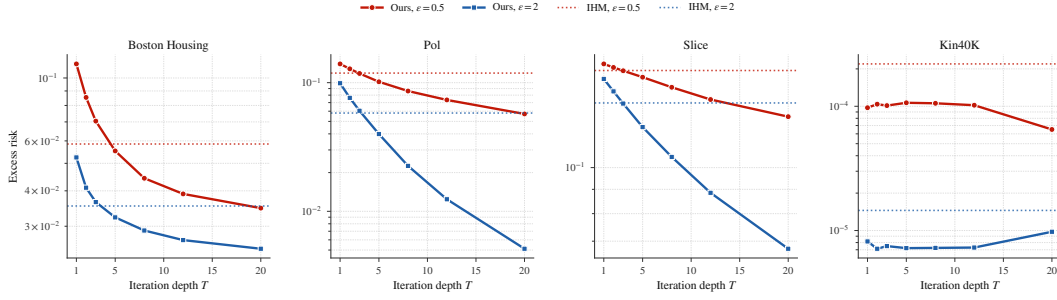


Figure 6: Excess risk vs. iteration depth T at fixed $\epsilon \in \{0.5, 2\}$ (dotted lines: IHM at $T=3$). Decoupling the single-shot sketch from T enables monotonic improvement to $T=20$ on convergence-limited datasets, while well-conditioned Kin40K exhibits a genuine interior optimum.

and 198 privacy configurations, DP-EBS-BB achieves state-of-the-art utility. A promising direction is extending single-shot preconditioning to generalized linear models: for logistic regression $w_{ii}(\theta) = p_i(1 - p_i) \leq 1/4$ gives $H(\theta) \preceq \frac{1}{4n} X^\top X$, so the single-shot preconditioner acts as a valid majorizer under majorization-minimization, guaranteeing monotone descent without T -fold composition.

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A COMPLETE DATASET CHARACTERISTICS

Table 3 lists the characteristics and configurations for all 33 UCI datasets.

Table 3: Characteristics and parameters for the complete 33-dataset UCI benchmark suite.

DATASET	n	d	$\lambda_{\min}(X^\top X)$	$\frac{1}{n}\ Y - X\theta^*\ _2^2$	SKETCH k	SPLIT (SK./GR.)
BOSTON HOUSING	404	13	2.478E-6	0.009585	100	0.49 / 0.51
TECATOR	192	124	-8.055E-16	0.000018	744	0.41 / 0.59
WINE	1087	11	7.536E-6	0.006790	112	0.52 / 0.48
BIKE SHARING	8708	18	0.000E+0	0.020990	137	0.56 / 0.44
AUTOMPG	313	7	3.274E-6	0.036460	97	0.50 / 0.50
ENERGY	614	9	1.097E-13	0.001599	105	0.51 / 0.49
CONCRETE	824	8	0.004981	0.015580	109	0.52 / 0.48
ELEVATORS	14939	18	3.945E-12	0.008325	144	0.58 / 0.42
GAS	2051	128	8.919E-13	0.006649	768	0.47 / 0.53
COMMUNITIES & CRIME	657	99	0.000454	0.039840	594	0.45 / 0.55
CONCRETE SLUMP	93	7	0.526100	0.001992	83	0.47 / 0.53
AIRFOIL	1202	5	2.102E-10	0.046380	113	0.54 / 0.46
BREAST CANCER	155	33	3.509E-19	0.039540	198	0.44 / 0.56
PARKINSONS	4700	21	5.599E-13	0.003684	130	0.54 / 0.46
SML	3724	26	0.000E+0	0.007136	156	0.53 / 0.47
KEGG (UNDIRECTED)	57248	27	-2.891E-17	0.001924	162	0.60 / 0.40
POL	13500	26	0.004879	0.183300	156	0.56 / 0.44
SOLAR	959	10	0.000E+0	0.010190	111	0.52 / 0.48
KEGG (DIRECTED)	43944	20	1.600E-16	0.008660	157	0.60 / 0.40
YACHT	277	6	0.173600	0.003540	96	0.50 / 0.50
SERVO	150	4	2.097000	0.069220	88	0.49 / 0.51
AUTOS	144	25	0.000E+0	0.009391	150	0.45 / 0.55
FERTILITY	90	9	1.639000	0.072240	82	0.46 / 0.54
FOREST	466	12	0.079580	0.058470	102	0.50 / 0.50
MACHINE	188	7	0.160200	0.022230	91	0.49 / 0.51
PROTEIN	41157	9	0.021770	0.119200	156	0.60 / 0.40
PENDULUM	567	9	4.349000	0.016580	104	0.51 / 0.49
SLICE	48150	385	-2.384E-16	0.026620	2310	0.53 / 0.47
BUZZ	524925	77	1.068E-6	0.028550	462	0.60 / 0.40
PUMADYN32NM	7372	32	126.400000	0.093610	192	0.54 / 0.46
3D ROAD	391387	3	13140.00	0.026460	183	0.60 / 0.40
KIN40K	36000	8	1911.000	0.063390	154	0.60 / 0.40
TAMI ELECTRIC	36624	3	32.14000	0.333800	154	0.60 / 0.40

B PRIVACY PROOFS AND COMPOSITION ACCOUNTING

We provide the formal privacy proof using RDP (Mironov, 2017) and analytic Gaussian mechanism calibration (Balle & Wang, 2018).

Definition 1 (Rényi Differential Privacy). *A mechanism $\mathcal{M}$ is (α, ϵ_{RDP}) -RDP if for all adjacent D, D' the Rényi divergence of order $\alpha > 1$ satisfies $D_\alpha(\mathcal{M}(D) \parallel \mathcal{M}(D')) \leq \epsilon_{RDP}$.*

Theorem 2. *Algorithm 1 is (ϵ, δ) -DP.*

Proof. Set $\epsilon_{\text{sketch}} = \text{sketch_frac} \cdot \epsilon$, $\epsilon_{\text{grad}} = \text{grad_frac} \cdot \epsilon$, $\delta_{\text{sketch}} = \delta_{\text{grad}} = \delta/2$. (1) *Sketch.* $\tilde{X}_0 = S_0 X + \eta \xi_0$; calibrating η by the analytic Gaussian mechanism (Lev et al., 2025) makes $\tilde{X}_0$ satisfy $(\epsilon_{\text{sketch}}, \delta_{\text{sketch}})$ -DP. (2) *Gradients.* At each step, Huber sensitivity is bounded by δ_t ; adding Gaussian noise with scale $\sigma_{\text{grad}} = \delta_t \sqrt{2 \ln(1.25/\delta_{\text{step}})}/\epsilon_{\text{step}}$ gives $(\epsilon_{\text{grad}}/T, \delta_{\text{grad}}/T)$ -DP per step, composing to $(\epsilon_{\text{grad}}, \delta_{\text{grad}})$ -DP by RDP. (3) *Post-processing.* $P^{-1}, \delta_t, \alpha_t^{\text{BB}}, \bar{\theta}_T$ depend only on $\tilde{X}_0$ and $\{\tilde{G}_t\}$, so cost zero budget. (4) *Composition.* $(\epsilon_{\text{sketch}} + \epsilon_{\text{grad}}, \delta_{\text{sketch}} + \delta_{\text{grad}}) = (\epsilon, \delta)$ -DP. $\square$

C EXCESS EMPIRICAL RISK BOUND

We sketch the derivation of the $\tilde{\mathcal{O}}(\frac{d}{n\epsilon} + \frac{\sqrt{d}}{n\epsilon})$ bound. Let $H = \frac{1}{n}X^\top X$ and $\bar{\sigma}_0^2 = \eta^2/(kn)$.

Lemma 2 (Sketch concentration). *For $k = 6 \max\{d, \log(4T/\varrho)\}$, with probability $\geq 1 - \varrho/2$, $(1 - \epsilon_{proj})H - \bar{\sigma}_0^2 I_d \preceq \frac{1}{n} \tilde{H}_0 \preceq (1 + \epsilon_{proj})H + 2\bar{\sigma}_0^2 I_d$, with $\epsilon_{proj} = \mathcal{O}(\sqrt{(d + \log(1/\varrho))/k})$.*

Proof sketch. Johnson-Lindenstrauss sub-Gaussian concentration (Woodruff, 2014; Pilanci & Wainwright, 2014) controls $\|\frac{1}{k} X^\top S_0^\top S_0 X - H\|_2 \leq \epsilon_{proj} \|H\|_2$; Marchenko-Pastur bounds control the noise term $\|\frac{\eta^2}{kn} \xi_0^\top \xi_0 - \bar{\sigma}_0^2 I_d\|_2 = \mathcal{O}(\bar{\sigma}_0^2 \sqrt{d/k})$. $\square$

Lemma 3 (Condition-number-free contraction). *The EB preconditioner yields $\|I_d - \alpha_t^{BB} P^{-1} H\|_2 \leq 1 - \rho$ with $\rho = \frac{1}{2(1+\epsilon_{proj})}$, independent of $\kappa(H)$.*

Combining a geometric decay $(1 - \rho)^{2T} \|\theta_0 - \theta^*\|_2^2 = \mathcal{O}(1/n^2)$ for $T = \mathcal{O}(\log n)$ with the PR-averaged private-gradient variance $\mathcal{O}(d/(n\epsilon_{grad}))$ and single-shot sketch error $\mathcal{O}(\sqrt{d}/(n\epsilon_{sketch}))$ yields $\frac{1}{2n} \|Y - X\bar{\theta}_T\|_2^2 - \frac{1}{2n} \|Y - X\theta^*\|_2^2 \leq \tilde{\mathcal{O}}(\frac{d}{n\epsilon} + \frac{\sqrt{d}}{n\epsilon})$ with probability $\geq 1 - \delta$.

D EXTENDED NUMERICAL RESULTS AND RUNTIME

Table 4 reports additional datasets, and Table 5 reports wall-clock runtime.

Table 4: Extended normalized excess empirical train risk (mean $\pm$ 95% CI) on additional datasets.

DATASET	METHOD	$\epsilon = 0.1$	$\epsilon = 0.631$	$\epsilon = 1.58$	$\epsilon = 10.0$
WINE ($n=1087, d=11$)	ADASSP	4.801E-1 $\pm$ 1.4E-3	4.090E-1 $\pm$ 1.2E-3	3.319E-1 $\pm$ 1.0E-3	1.841E-1 $\pm$ 5.1E-4
	LINMIX	4.650E-1 $\pm$ 4.7E-3	3.593E-1 $\pm$ 5.3E-3	2.705E-1 $\pm$ 4.7E-3	1.642E-1 $\pm$ 2.3E-3
	IHM	4.565E-1 $\pm$ 2.1E-3	3.158E-1 $\pm$ 1.7E-3	2.146E-1 $\pm$ 1.0E-3	1.323E-1 $\pm$ 6.8E-4
	DP-EBS-BB	4.604E-1 $\pm$ 1.6E-3	2.729E-1 $\pm$ 2.0E-3	1.848E-1 $\pm$ 1.3E-3	1.173E-1 $\pm$ 9.5E-4
COMMUNITIES & CRIME ($n=657, d=99$)	ADASSP	1.749E-1 $\pm$ 2.4E-4	1.699E-1 $\pm$ 2.3E-4	1.624E-1 $\pm$ 2.1E-4	1.215E-1 $\pm$ 1.8E-4
	LINMIX	1.739E-1 $\pm$ 8.9E-4	1.624E-1 $\pm$ 1.1E-3	1.428E-1 $\pm$ 1.0E-3	8.220E-2 $\pm$ 9.0E-4
	IHM	1.724E-1 $\pm$ 6.1E-4	1.555E-1 $\pm$ 5.6E-4	1.312E-1 $\pm$ 5.3E-4	5.500E-2 $\pm$ 2.1E-4
	DP-EBS-BB	1.730E-1 $\pm$ 3.2E-4	1.580E-1 $\pm$ 3.3E-4	1.317E-1 $\pm$ 3.8E-4	4.777E-2 $\pm$ 2.3E-4
PARKINSONS ($n=4700, d=21$)	ADASSP	5.704E-2 $\pm$ 6.0E-4	2.432E-2 $\pm$ 1.3E-4	1.789E-2 $\pm$ 6.4E-5	1.254E-2 $\pm$ 1.6E-5
	LINMIX	4.018E-2 $\pm$ 1.6E-3	2.077E-2 $\pm$ 5.2E-4	1.640E-2 $\pm$ 3.1E-4	1.350E-2 $\pm$ 1.8E-4
	IHM	2.467E-2 $\pm$ 4.6E-4	1.498E-2 $\pm$ 1.4E-4	1.262E-2 $\pm$ 6.8E-5	1.098E-2 $\pm$ 7.2E-5
	DP-EBS-BB	2.107E-2 $\pm$ 2.2E-4	1.399E-2 $\pm$ 1.3E-4	1.239E-2 $\pm$ 7.4E-5	1.032E-2 $\pm$ 5.9E-5
PROTEIN ($n=41157, d=9$)	ADASSP	4.712E-2 $\pm$ 4.2E-5	4.492E-2 $\pm$ 4.5E-5	4.173E-2 $\pm$ 4.8E-5	2.730E-2 $\pm$ 7.8E-5
	LINMIX	4.732E-2 $\pm$ 2.2E-4	4.472E-2 $\pm$ 4.9E-4	4.134E-2 $\pm$ 7.0E-4	2.925E-2 $\pm$ 8.0E-4
	IHM	4.657E-2 $\pm$ 6.2E-5	4.200E-2 $\pm$ 6.5E-5	3.572E-2 $\pm$ 7.6E-5	1.668E-2 $\pm$ 7.6E-5
	DP-EBS-BB	4.590E-2 $\pm$ 1.3E-4	3.838E-2 $\pm$ 1.4E-4	2.989E-2 $\pm$ 1.4E-4	1.277E-2 $\pm$ 1.1E-4

Table 5: Wall-clock runtime (seconds per trial) across algorithms on representative datasets.

DATASET (n, d)	DP-SGD ($P=I$)	ADASSP	IHM ($T=3$)	DP-EBS-BB
BOSTON HOUSING (404, 13)	0.008	0.004	0.024	0.012
PARKINSONS (4700, 21)	0.035	0.012	0.118	0.048
PROTEIN (41157, 9)	0.182	0.042	0.450	0.210
SLICE (48150, 385)	0.280	0.110	1.180	0.420
BUZZ (524925, 77)	1.420	0.350	3.850	1.680